

■ Thread

`Thread[f[args]]` “threads” f over any lists that appear in $args$.

`Thread[f[args], n]` threads f over lists that appear in the first n $args$.

`Thread[f[args], -n]` threads over the last n $args$.

`Thread[f[args], {n, m}]` threads over arguments n through m .

`Thread[f[args], spec, h]` threads f over any objects with head h that appear.

Example: `Thread[f[{a,b}, c, {d,e}]]` $\longrightarrow$ `{f[a, c, d], f[b, c, e]}`. ■ Functions with attribute `Listable` are automatically threaded over lists. ■ All the elements in $args$ whose head is h must be of the same length. ■ Arguments that do not have head h are copied as many times as there are elements in the arguments that do have head h . ■ `Thread` uses the standard *sequence specification*. ■ See page 250. ■ See also: `Distribute`, `Map`, `Inner`.